Solutions

1. **Solution:** $V = \ell wh \Rightarrow 864 = 12 \cdot 6 \cdot h \Rightarrow h = 12$ cm.
$SA = 2(\ell w + \ell h + wh) = 2(72 + 144 + 72) = 576$ cm^2.

2. **Solution:** Open-top area: bottom $= \ell w = 216$, sides $= 2\ell h + 2wh = 2(18 \cdot 10) + 2(12 \cdot 10) = 360 + 240 = 600$. Total $= 816$ cm^2.
Ribbon loop length equals perimeter of cross-section around the sides: $2(\ell + w) = 2(18 + 12) = 60$ cm.

3. **Solution:** Equal volumes: $\pi(4)^2 h = \frac{1}{3}\pi(4)^2(18) \Rightarrow h = 6$ cm.
Lateral areas: cylinder $= 2\pi rh = 2\pi(4)(6) = 48\pi$; cone $= \pi rs$ with $s = \sqrt{18^2 + 4^2} = \sqrt{340} > \sqrt{324} = 18$. Since r same, cone's lateral area $\pi \cdot 4 \cdot s > 72\pi$ which is $> 48\pi$. So cone has greater lateral area.

4. **Solution:** (a) $h = \sqrt{s^2 - r^2} = \sqrt{10^2 - 6^2} = \sqrt{64} = 8$ cm.
(b) $SA_T = \pi r(r + s) = \pi(6)(6 + 10) = 96\pi$ cm^2.
(c) $V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(36)(8) = 96\pi$ cm^3.

5. **Solution:** Label area =(circumference)×(label height)= $(2\pi r) \cdot 8 = 264$. So $r = \dfrac{264}{16\pi} = \dfrac{33}{2\pi} \approx 5.25$ cm. The cylinder's height equals label height $\Rightarrow h = 8$ cm.

6. **Solution:** $4\pi r^2 = 324\pi \Rightarrow r^2 = 81 \Rightarrow r = 9$ cm. Volume $= \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(729) = 972\pi$ cm^3.

7. **Solution:** (a) $SA = a^2 + 2as = 14^2 + 2(14)(13) = 196 + 364 = 560$ cm^2.
(b) $h = \sqrt{s^2 - (a/2)^2} = \sqrt{13^2 - 7^2} = \sqrt{169 - 49} = \sqrt{120} = 2\sqrt{30}$.
$V = \frac{1}{3}a^2 h = \frac{1}{3}(196)(2\sqrt{30}) = \frac{392}{3}\sqrt{30}$ cm^3.

8. **Solution:** Same volume 960π: $h = \dfrac{960}{r^2}$. So $h_A = \dfrac{960}{36} = 26.\overline{6}$ cm; $h_B = \dfrac{960}{64} = 15$ cm.
Lateral area $2\pi rh = 2\pi \cdot r \cdot \dfrac{960}{r^2} = \dfrac{1920\pi}{r}$, which is *smaller* for larger r. So Can B uses less lateral area.

9. **Solution:** $V_{\text{cone}} = \frac{1}{3}\pi(5^2)(12) = 100\pi$. Cylinder radius same $r = 5$, so $\pi r^2 H = 100\pi \Rightarrow 25H = 100 \Rightarrow H = 4$ cm.

10. **Solution:** Base $A = \ell w = 144 \Rightarrow \ell = \dfrac{144}{w} = \dfrac{144}{8} = 18$ cm.
$SA = 2(\ell w + \ell h + wh) = 2(144 + 18 \cdot 15 + 8 \cdot 15) = 2(144 + 270 + 120) = 2 \cdot 534 = 1068\ \text{cm}^2$.

11. **Solution:** Outside area of closed cylinder: $SA = 2\pi r(h+r) = 2\pi(3)(5+3) = 48\pi \approx 150.8\ \text{m}^2$.
Gallons $= \dfrac{150.8}{28} \approx 5.39 \Rightarrow$ need **6** gallons.

12. **Solution:** (a) $V = \pi r^2 h + \frac{2}{3}\pi r^3 = \pi(6^2)(10) + \frac{2}{3}\pi(6^3) = 360\pi + 144\pi = 504\pi\ \text{cm}^3$.
(b) Exterior area: curved cylinder $= 2\pi rh = 2\pi(6)(10) = 120\pi$; hemisphere area $= 2\pi r^2 = 72\pi$; bottom circle $= \pi r^2 = 36\pi$. Total $= 228\pi\ \text{cm}^2$.

13. **Solution:** Cube side s from $s^3 = 512 \Rightarrow s = 8$ cm. Sphere radius $r = s/2 = 4$ cm.
$SA = 4\pi r^2 = 64\pi\ \text{cm}^2$; $V = \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(64) = \frac{256}{3}\pi\ \text{cm}^3$.

14. **Solution:** $V_{\text{cone}} = \frac{1}{3}Bh$, $V_{\text{pyr}} = \frac{1}{3}Bh$ as well. If cone volume is 120, then pyramid volume is also 120 $\mathbf{cm}^3$ (same B and h, same $\frac{1}{3}Bh$ structure).

15. **Solution:** (a) Volume $= 0.80 \cdot 0.50 \cdot 0.45$ m^3 $= 0.18$ m^3 $= 180{,}000$ cm^3 $= 180$ L.
(b) $0.85 \times 180 = 153$ L.

16. **Solution:** (a) Grid 3×2 fits exactly (diameter 10 spans height 10): **6 balls**.
(b) Box volume $= 30 \cdot 20 \cdot 10 = 6000$ cm^3. One sphere $V = \frac{4}{3}\pi(5^3) = \frac{500}{3}\pi$. Six spheres $= 1000\pi \approx 3141.6$ cm^3. Fraction $\approx \dfrac{1000\pi}{6000} = \dfrac{\pi}{6} \approx 0.524$.

17. **Solution:** Cylinder $V = \pi r^2 h = \pi r^3$ (since $h = r$). Sphere $V = \frac{4}{3}\pi r^3$. Thus sphere has greater volume by factor $\frac{4}{3}$.

18. **Solution:** Equal surface areas: cylinder $SA = 2\pi r(h + r) = 2\pi r(2r + r) = 6\pi r^2$. Sphere $4\pi R^2$.
Set $4\pi R^2 = 6\pi r^2 \Rightarrow R^2 = \frac{3}{2}r^2 \Rightarrow R = \sqrt{\frac{3}{2}}\,r$.